Band-1 Andrews–Curtis: AK(2) Sequence Replay

Collapse 2.33 next_wave #1 — methods / verification replay only (box BFS + Havas–Ramsay lit cross-check)

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-14

Document class. Scientific verification memo (Band-1; collapse **2.33**).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parent. `2026-09-11_Band1_Andrews_Curtis_presentation_probe` (toys certified; shallow AK(2) miss; AK(3) OPEN).
Lab. `/workspace/experiments/band1/andrews_curtis_ak2_replay/`
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is **methods / verification replay only**. It **does not** stamp Clay-closed. It **does not** claim the Andrews–Curtis conjecture solved. It **does not** claim AK(3) trivialized or disproved. It **does not** claim a finished Theory of Everything. GATE-04 collection remains **LOCKED** (no invented meter J). No RH / P≠NP / YM "solved" claims. Non-finding on AK(3) $\neq$ counterexample.

Abstract

Prior Band-1 probe (2026-09-11) AC-trivialized toy presentations $\langle a, b \mid ab, b\rangle$ and $\langle a, b \mid ab, a\rangle$, but a shallow BFS on the Akbulut–Kirby presentation

$$\mathrm{AK}(2) = \langle a, b \mid a^2 b^{-3},\ abab^{-1}a^{-1}b^{-1}\rangle$$

missed a known trivialization (literature: AK(2) is AC-trivializable; intermediate total length ≤ 12 is impossible). This note **replays** an explicit AC1–AC3 sequence on box: a **17-move** path rediscovered by deeper BFS (raised `max_total_len=15`), verified end-to-end to the canonical trivial presentation $\langle a, b \mid a, b\rangle$, plus an independent **14-move** lit cross-check from Havas–Ramsay (2003) on an AC-equivalent start. A bounded non-finding on AK(3) is re-ledgered **OPEN**.

Keywords. Andrews–Curtis; Akbulut–Kirby AK(2); AC1–AC3; verification replay; Havas–Ramsay; Band-1; non-claim.

1. Setup and moves

Generators. Free group $F_2 = \langle a, b\rangle$. Relators encoded as integer words ($a \mapsto +1$, $b \mapsto +2$, inverses negative), freely reduced after every move.

Elementary AC moves (box implementation).

Tag	Action
AC1	$r_i := r_i\, r_j^{\pm 1}$ for $i \neq j$
AC2	$r_i := r_i^{-1}$
AC3	$r_i := g^{-1} r_i g$ with $\lvert g\rvert = 1$ (generator or inverse)

Canonical target. A presentation is **trivial** here iff both relators are unit-length and hit $\{a^{\pm 1}, b^{\pm 1}\}$ exactly once each (order and inverses free). Notation in tables also uses Havas capitals ($A = a^{-1}$, $B = b^{-1}$).

Family. $\mathrm{AK}(n) = \langle a, b \mid a^n b^{-(n+1)},\ abab^{-1}a^{-1}b^{-1}\rangle$. Here $n = 2$ (length 11) is the replay target; $n = 3$ (length 13) remains the named open ledger.

2. Why the 2026-09-11 shallow probe missed

Havas–Ramsay prove that **no** AC-proof of AK(2) exists if intermediate presentations are limited to total relator length ≤ 12. Our prior probe used `max_rel_len=12`, `max_depth=8`, `max_states=40\,000` — consistent with a miss. Raising the total-length bound to **15** (the Havas length that admits a 14-move essential proof) yields box rediscovery.

3. Primary result: box BFS rediscovery (PASS)

Source label. `box BFS found` (primary).
Bounds. `max_total_len=15`, `max_rel_len=15`, `max_depth=20`, `max_states=800\,000`.
Outcome. Status `TRIVIALIZED`; **17** moves; visited **40141**; wall ≈ **2.21 s**; seen **58699**.
Verifier. Every labeled move re-applied from the canonical start; final presentation $\langle a, b \mid a, b\rangle$; **PASS**.
Max intermediate total length. 14 (within the raised bound).

3.1 Step table (box 17-move sequence)

Step	Move	Havas form	Tot. len
0	START	($aaBBB$, $abaBAB$)	11
1	AC3: conj r_2 by b	($aaBBB$, $BabaBA$)	11
2	AC1: $r_2{*}{=}r_1$	($aaBBB$, $BabaBaBBB$)	14
3	AC3: conj r_1 by a	($aBBBa$, $BabaBaBBB$)	14
4	AC3: conj r_2 by B	($aBBBa$, $abaBaBBBB$)	14
5	AC3: conj r_2 by a	($aBBBa$, $baBaBBBBa$)	14
6	AC1: $r_2{*}{=}r_1^{-1}$	($aBBBa$, $baBaBA$)	11
7	AC3: conj r_2 by a	($aBBBa$, $AbaBaB$)	11
8	AC1: $r_1{*}{=}r_2$	($aBBaBaB$, $AbaBaB$)	13
9	AC3: conj r_1 by a	($BBaBaBa$, $AbaBaB$)	13
10	AC3: conj r_1 by B	($BaBaBaB$, $AbaBaB$)	13
11	AC1: $r_2{*}{=}r_1^{-1}$	($BaBaBaB$, $AbbAb$)	12
12	AC3: conj r_1 by B	($aBaBaBB$, $AbbAb$)	12
13	AC3: conj r_2 by A	($aBaBaBB$, $bbAbA$)	12
14	AC1: $r_1{*}{=}r_2$	(aB, $bbAbA$)	7
15	AC1: $r_2{*}{=}r_1$	(aB, bbA)	5
16	AC1: $r_2{*}{=}r_1$	(aB, b)	3
17	AC1: $r_1{*}{=}r_2$	(a, b)	2

Certified. End state is the canonical trivial presentation (PASS).

4. Lit cross-check: Havas–Ramsay 2003 (PASS)

Citation. G. Havas & C. Ramsay, *Breadth-first search and the Andrews–Curtis conjecture*, Internat. J. Algebra Comput. **13**(1) (2003) 61–68, §3.

Start. ($aBBBa$, $BAbabA$) — AC-equivalent to canonical AK(2)($aaBBB$, $abaBAB$) via cyclic conjugation / invert (setup moves not counted in their "14 essential").

Published chain (14 essential moves; length-15 bound). Verified on box by inferring each

single AC1–AC3 neighbor and replaying; final $(B,\ A)$ — trivial (inverses of generators). **PASS.** Havas–Ramsay also prove shortest essential length ≥ 14 (bidirectional unconstrained search).

Step	Inferred move	From → To (Havas)	Tot. len
1	AC1: $r_2{*}{=}r_1$	$(aBBBa, BAbabA) \to (aBBBa, BAbaBBa)$	12
2	AC3: conj r_2 by B	$\to (aBBBa, AbaBBaB)$	12
3	AC1: $r_1{*}{=}r_2$	$\to (aBBaBBaB, AbaBBaB)$	15
4	AC2: invert r_2	$\to (aBBaBBaB, bAbbABa)$	15
5	AC3: conj r_2 by A	$\to (aBBaBBaB, abAbbAB)$	15
6	AC3: conj r_2 by b	$\to (aBBaBBaB, BabAbbA)$	15
7	AC1: $r_2{*}{=}r_1$	$\to (aBBaBBaB, BaBaB)$	13
8	AC3: conj r_1 by a	$\to (BBaBBaBa, BaBaB)$	13
9	AC2: invert r_2	$\to (BBaBBaBa, bAbAb)$	13
10	AC3: conj r_2 by b	$\to (BBaBBaBa, AbAbb)$	13
11	AC1: $r_1{*}{=}r_2$	$\to (BBa, AbAbb)$	8
12	AC1: $r_2{*}{=}r_1$	$\to (BBa, Ab)$	5
13	AC1: $r_1{*}{=}r_2$	$\to (B, Ab)$	3
14	AC1: $r_2{*}{=}r_1$	$\to (B, A)$	2

Earlier lit. Miasnikov (1999) first established AK(2) AC-trivializability via genetic search (longer chain; grows to length 25). Swan–Ochoa–Kendall–Edjvet (2012) report a length-10 path to a one-generator intermediate under their forward success criterion — not re-claimed as a full two-relator trivialization here.

5. AK(3) OPEN ledger (re-run)

$$\mathrm{AK}(3) = \langle a, b \mid a^3 b^{-4},\ abab^{-1}a^{-1}b^{-1} \rangle$$

(total length 13). Bounded box BFS this run: `max_total_len=18`, `max_rel_len=18`, `max_depth=12`, `max_states=200\,000` ⇒ **NOT_FOUND_IN_BOUNDS** (visited **11176**, wall ≈ **0.50 s**, best length stuck at 13).

Ledger. Ordinary AC-trivialization of AK(3) remains **OPEN**. Bounded non-finding ≠ counterexample ≠ disproof. Stable-AC literature is nuanced/contested and is **not** stamped here. AK(3) is still the shortest (by total relator length) widely cited potential counterexample class for ordinary AC.

6. Non-claims (explicit)

Claim	Status in this memo
Andrews–Curtis conjecture solved	**NO**
AK(2) is a new mathematical discovery	**NO** (lit-known; verification replay)
AK(3) trivialized	**NO** — OPEN
AK(3) disproved / counterexample	**NO**
Clay Millennium stamp	**NO**
Finished ToE / GATE-04 collection	**NO** — GATE-04 **LOCKED**
RH / P≠NP / YM solved	**NO**

7. Artifacts and reproducibility

Path	Role
`experiments/band1/andrews_curtis_ak2_replay/replay_ak2.py`	BFS + verify + Havas replay
`.../results.json`	Machine-readable PASS/OPEN ledger
`.../SUMMARY.txt`	One-page human summary
`.../literature_ledger.json`	Citations + status entries
`papers/2026-09-14_Band1_Andrews_Curtis_AK2_Sequence_replay.md`	This memo

Reproduce. `python3 replay_ak2.py` on box; expect AK(2) box verify PASS, Havas verify PASS, AK(3) NOT_FOUND / OPEN.

8. Next optional (honest)

1. Import / verify a public length-14 Miller–Schupp / Carreras certificate (verification only).

2. Do **not** claim AK(3) or full AC closed.

3. GATE-04 / Track B privacy and Object 1 meter remain blocking for collection — orthogonal to this Band-1 replay.

References (selected)

1. J.J. Andrews & M.L. Curtis, Free groups and handlebodies, *Proc. Amer. Math. Soc.* **16** (1965) 192–195.

2. S. Akbulut & R. Kirby, A potential smooth counterexample…, *Topology* **24** (1985) 375–390.

3. A.D. Miasnikov, Genetic algorithms and the Andrews–Curtis conjecture, *Internat. J. Algebra Comput.* **9** (1999) 671–686.

4. G. Havas & C. Ramsay, Breadth-first search and the Andrews–Curtis conjecture, *Internat. J. Algebra Comput.* **13**(1) (2003) 61–68.

5. J. Swan, G. Ochoa, G. Kendall & M. Edjvet, Fitness landscapes and the Andrews–Curtis conjecture, *Internat. J. Algebra Comput.* **22**(2) (2012) 1250009.

6. A. Shehper et al., What makes math problems hard for reinforcement learning. . . , arXiv:2408.15332.

End stamp. Methods / verification replay · AK(2) PASS (box 17 + Havas 14) · AK(3) OPEN · AC **not** solved · Clay **not** stamped · GATE-04 **LOCKED** · 2026-09-14.